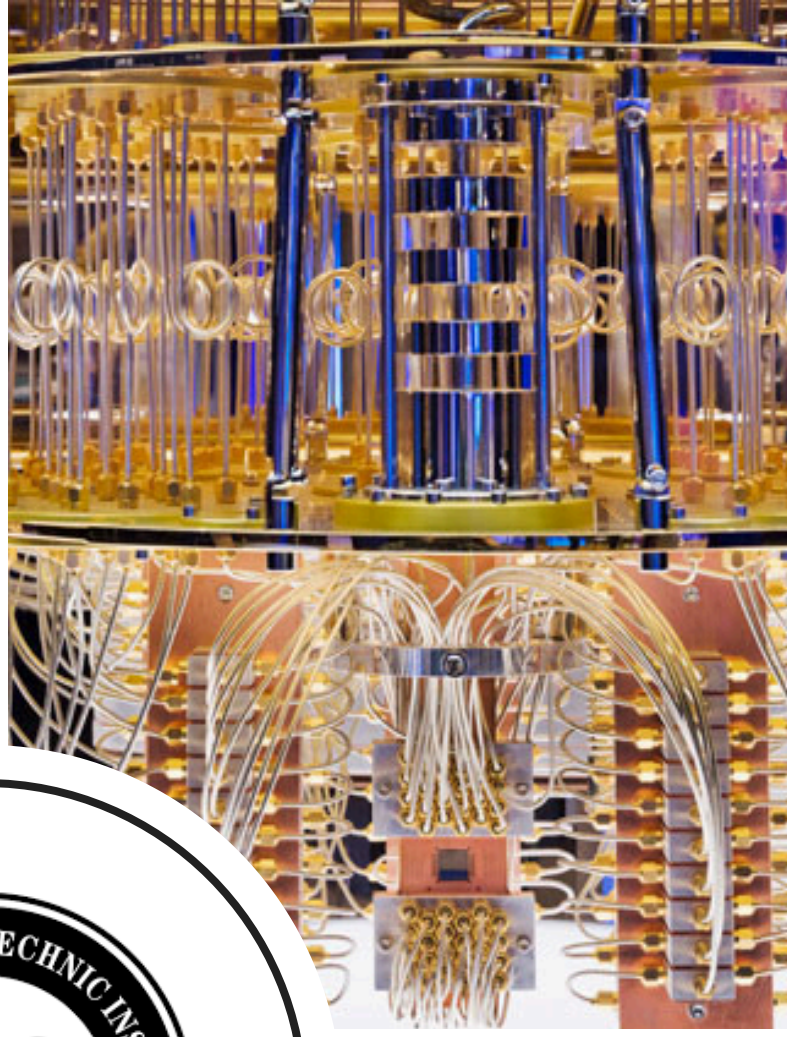




Graduate Program Guide

2026 - 2027





Computer Science Department Graduate Program Guide 2026-2027

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1. Introduction

This Graduate Program Guide describes the academic requirements, policies, and milestones for the Master of Science and Doctor of Philosophy programs in Computer Science at Rensselaer Polytechnic Institute. It is intended to help students understand degree requirements, plan their programs of study, and successfully progress toward graduation.

Students are responsible for meeting the requirements described in this guide as well as the policies of the Office of Graduate Education and the Institute. Questions about degree requirements, policies, or procedures should be directed to the appropriate graduate program contact.

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2. MS Program

The Master of Science degree in Computer Science at RPI is a technical degree from which students may advance to positions of responsibility in the computing field with a solid foundation of knowledge to serve them. A number of students will continue into PhD study and will be similarly well prepared. An MS degree provides considerable flexibility for management and leadership positions within the private and public sectors. RPI's MS students have benefitted from both the strong problem-solving orientation of CS faculty researchers as well as the culture of technology and application of the institution and within the curriculum.

2.1. MS Degree Requirements

In addition to meeting the degree requirements of the Office of Graduate Education, a candidate must:

- Complete 30 credits that meet the following requirements:
  - Courses must be 4000 or 6000-level courses. A 4000-level course may be applied to the MS degree only if the course is not offered at the 6000 level.
  - At least 18 credits must be at the 6000 level.
  - At least 15 credits must be Computer Science courses (CSCI prefix).
  - No more than 6 credits may be independent study credits.
  - Include the 1-credit Graduate Skills course (“CSCI 69xx Graduate Skills”), which must be taken in the first fall semester of enrollment.

- Complete the Core Foundation and Core Depth and Breadth requirements described below.
- Maintain a minimum overall GPA of 3.0 (B average) in all courses applied toward the degree. Students who receive two failing grades will be dismissed from the program.
- Attend at least 50% of department colloquia each semester while enrolled as a full-time student (up to four semesters). See the Colloquium Attendance section for details.
- Complete either a Master's Thesis or a Master's Project.

All MS students must complete a Plan of Study form by the second semester of study. For students in the Accelerated Master's program, the Plan of Study is completed during the application process. This plan is drawn up with the advice and approval of the student's academic advisor and is to be a coherent, thoughtful plan reflecting the student's professional goals. If necessary, changes can be made to this plan at any time with the approval of the academic advisor and the graduate program director. The Plan of Study must remain up to date throughout the student's program.

2.1.1. Core Foundation Requirements

Students must demonstrate advanced undergraduate-level competence in the following core areas:

1. Algorithms
2. Systems
3. Software/Programming Languages

Who Must Complete Which Areas

- Students with a BS in Computer Science from RPI automatically satisfy the Core Foundation requirement.
- Students with a BS in Computer Science from another institution must demonstrate competence in at least two of the three areas.
- All other students must demonstrate competence in all three areas unless granted an exception by the Graduate Curriculum Committee (GCC).

Students without a BS in Computer Science may petition the GCC to satisfy only two of the three areas. Students without an approved petition must complete all three areas.

Methods for Demonstrating Competence

Students may satisfy a core area in one of the following ways:

1. Earn a grade of B or better in the corresponding undergraduate course:
   - Algorithms → "Design and Analysis of Algorithms"

- Systems → “Operating Systems”, “Advanced Operating Systems”, or “Advanced Compilers”
- Software/Programming Languages → “Programming Languages”

2. Earn a grade of B or better on the final exam for the corresponding undergraduate course. Students pursuing this option should coordinate with the Graduate Program Administrator and are encouraged to audit the course.
3. Complete an approved graduate-level course in the corresponding area (marked with *, **, or *** in the Course Lists for the Core Buckets)

Students without an undergraduate degree in Computer Science may satisfy a Core Foundation area through Option 3 only with **prior GCC approval**. Approval must be obtained before enrolling in the graduate-level course. Requests should include an official transcript and course syllabus demonstrating preparation equivalent to the corresponding undergraduate core course.

2.1.2. Core Depth and Breadth Requirements

Students must take 4 6000-level courses from the Core Course Lists:

- **Depth:** Two courses must be in one Core Bucket.
- **Breadth:** At least one course must be from a different Core Bucket than the depth bucket
- Students must earn at least 12 total quality points across the four courses, based on the standard RPI grading scale (A = 4.0, A− = 3.67, B+ = 3.33, B = 3.0, B-=2.67, etc.).

2.1.3. MS Thesis Option

Students pursuing the thesis option must complete 6–9 credits of CSCI-6990 Master’s Thesis. A maximum of 9 thesis credits may be applied toward the degree.

The thesis must present an original research contribution and is supervised by a faculty advisor. Students may work with a research advisor outside the Computer Science Department, but the thesis must also be supervised by a Computer Science faculty advisor.

The thesis is evaluated by a committee of three faculty members, including the research advisor(s), and is graded on a satisfactory/unsatisfactory (S/U) basis. Committee membership must be approved by the Office of Graduate Education (OGE) prior to thesis submission. Students must submit the Nomination of Masters Thesis Committee form to the department’s Graduate Program Administrator who will manage the OGE approval process.

Students are encouraged to begin thesis research early in the program and may register for thesis credits over multiple semesters. Most thesis students register for at least one thesis credit each semester while conducting research.

Before graduation, students must:

- Submit the completed thesis to their advisor(s) for review at least one week before the OGE submission deadline
- Obtain advisor approval of the thesis content and format
- Submit the approved thesis to OGE by the posted deadline
- Complete an oral presentation to their faculty committee.

All thesis work must meet RPI standards of academic integrity and graduate scholarship.

2.1.4. MS Project Option

The project option requires at least three, and no more than four credits of project credits. Students completing a Master's project must register for CSCI-6980 Master's Project. CSCI-6980 is graded S or U.

Master's projects are completed under the guidance of a project advisor. The project advisor must be someone who has a permanent teaching appointment (tenure track or teaching faculty) to the Department of Computer Science. Students may also collaborate with a research supervisor who does not fit this description, as long as there is also a project advisor who does meet the criteria and bears ultimate responsibility for supervising the project.

The project scope will be agreed upon by both the advisor and the student. The workload for completing a project will be similar to the workload for a full semester advanced 3-4 credit course.

All Master's projects must be presented as posters at a Computer Science Department poster session. Poster sessions will be held in November and April, early enough to allow students to incorporate feedback from the poster session into their final projects.

2.2. MS Program: Quantum Computing Track

The Quantum Computing Track is available to students pursuing the MS in Computer Science who wish to develop specialized expertise in quantum computing, quantum algorithms, and related computational methods.

Students in the Quantum Computing Track must satisfy all requirements for the MS in Computer Science, except that the standard Core Depth and Breadth requirement is replaced by the Quantum Computing Track Depth and Breadth requirement described below.

2.2.1. Quantum Computing Track Depth and Breadth Requirement

Students must complete four courses with a grade of B or better that satisfy the following requirements:

- **Depth:** At least two courses from the Quantum Computing Depth Bucket.

- **Quantum Computing Breadth:** At least one additional course from either the Quantum Computing Depth Bucket or the Quantum Computing Supplementary Course List.
- **Computer Science Breadth:** At least one course from a CS Core Bucket.

2.2.2. Quantum Computing Project/Thesis Requirement

Students must complete one of the following:

- A 3-4 credit quantum computing-focused Master's Project (CSCI 6980);
- A 6-9 credit quantum computing-focused Master's Thesis (CSCI 6990); or
- An approved 3-4 credit project-based course in quantum computing.

3. PhD Program

The PhD in Computer Science is the highest professional degree awarded by the Rensselaer Department of Computer Science. With it, students may advance to university teaching and research, and to careers in industrial research, with a solid foundation of knowledge and an ability to carry through original investigations in Computer Science.

Students progress through the PhD program by completing the following milestones:

1. Core Qualifying Exam (by end of third and fourth semesters)
2. Research Qualifying Exam (by the end of the fourth semester)
3. Candidacy Exam (by end of fourth year / one year before defense)
4. Dissertation and Defense (when ready)

Students pursuing the MS and PhD concurrently are expected to make progress toward PhD milestones from the start of their graduate studies. Part-time students may take up to twice the time to meet these milestones.

Early in the program, each student must identify a primary advisor who will oversee the student's research and academic progress. The primary advisor must hold an appointment in the Department of Computer Science. Students may also have one or more co-advisors from other academic departments at RPI when appropriate for the student's research.

3.1. Program Requirements

To earn the PhD in Computer Science, students must satisfy the following program requirements:

- Complete 72 credits satisfying the Course and Credit Requirements described below.
- Pass the Core Qualifying Exam
- Pass the Research Qualifying Exam
- Complete and successfully defend a doctoral dissertation based on original research.

- Maintain a minimum GPA of 3.0 (B average).
- Attend at least 50% of department colloquia each semester while enrolled as a full-time student (up to four semesters). See the Colloquium Attendance section for details.
- Complete the 1-credit Graduate Skills course ("CSCI 69xx Graduate Skills"), which must be taken in the first fall semester of enrollment.
- Deliver at least one public technical presentation.

3.1.1. Course and Credit Requirements

Students Entering with a Bachelor's Degree

Students entering the PhD program with a bachelor's degree must complete **72 credits** that satisfy the following requirements:

- At least 36 credits must be course credits.
- Course credits must be at the 4000 or 6000 level. A 4000-level course may be applied to the PhD degree only if the course is not offered at the 6000 level.
- At least two-thirds of the course credits must be at the 6000 level.
- At least 24 credits must be dissertation credits (CSCI-9990).
- No more than 9 credits may be independent study credits.

Students Entering with a Master's Degree

Students entering the PhD program with a relevant Master's degree may apply **24 credits** from the Master's degree toward the PhD credit requirement. Such students must complete **48 additional credits** at RPI that satisfy the following requirements:

- At least 12 credits must be course credits.
- Course credits must be at the 4000 or 6000 level. A 4000-level course may be applied to the PhD degree only if the course is not offered at the 6000 level.
- At least 24 credits must be dissertation credits (CSCI-9990).
- No more than 6 credits may be independent study credits.

3.1.2. Core Qualifying Exam

The Core Qualifying Exam is the first major milestone in the PhD program. It is designed to ensure that students acquire both breadth in computer science and depth in their primary research area through a combination of coursework and competency requirements.

To pass the Core Qualifying Exam, students must:

- Complete the Core Foundation Requirement.
- Complete the Core Depth and Breadth Requirement.

Full-time students must complete the Core Depth and Breadth Requirement and two of the three Core Foundation areas by the end of the third semester. The remaining Core Foundation area must be completed by the end of the fourth semester. Part-time students may take up to six semesters to complete the Core Qualifying Exam requirements.

Milestone timing is measured from a student's first semester in the PhD program, regardless of concurrent enrollment in another degree program or prior enrollment in a different graduate program. For students in the accelerated BS–PhD program, milestone timing begins upon completion of 128 undergraduate credits.

Students should work closely with their advisor to develop a plan for satisfying the Core Qualifying Exam requirements. Once all requirements have been completed, students must submit the Core Qualifying Exam certification form, signed by both the student and advisor, to the Graduate Program Administrator.

Core Foundation Requirement

Students must demonstrate advanced undergraduate-level competence in the following core areas:

1. Algorithms
2. Systems
3. Software/Programming Languages

Students may satisfy a core area in one of the following ways:

1. Earn a grade of B or better in the corresponding undergraduate course:
    - Algorithms → "Design and Analysis of Algorithms"
    - Systems → "Operating Systems", "Advanced Operating Systems", or "Advanced Compilers"
    - Software/Programming Languages → "Programming Languages"
2. Earn a grade of B or better on the final exam for the corresponding undergraduate course. Students pursuing this option should coordinate with the Graduate Program Administrator and are encouraged to audit the course.
3. Complete an approved graduate-level course in the corresponding area (marked with *, **, or *** in the Core Bucket Course Lists).

Students without an undergraduate degree in Computer Science may satisfy a Core Foundation area through Option 3 only with **prior GCC approval**. Approval must be obtained before enrolling in the graduate-level course. Requests should include an official transcript and course syllabus demonstrating preparation equivalent to the corresponding undergraduate core course.

Core Depth and Breadth Requirement

Students must complete five 6000-level courses from the approved Core Bucket Course Lists.

- **Depth:** Three courses must be completed within a single bucket corresponding to the student's primary research area.
- **Breadth:** Two courses must be completed from two different buckets outside the depth bucket.
- Students must earn at least 18.5 total quality points across the five courses, based on the standard RPI grading scale (A = 4.0, A− = 3.67, B+ = 3.33, B = 3.0, etc.).

Courses used to satisfy the Core Foundation Requirement may also be used to satisfy the Core Depth and Breadth Requirement, provided they meet the requirements of both.

3.1.3. Research Qualifying Exam

The Research Qualifying Exam is the second major milestone in the PhD program. It is designed to ensure that students have developed a thorough understanding of their research area, including the relevant literature, major contributions, and open research questions.

To pass the Research Qualifying Exam, students must deliver a survey-style oral presentation before a faculty committee. The presentation should demonstrate the student's understanding of the broader research area in which their work is situated and place their research within the context of existing work.

Full-time students must complete the Research Qualifying Exam by the end of the fourth semester in the PhD program. Part-time students must complete the Research Qualifying Exam by the end of the sixth semester.

Students should work closely with their advisor and committee when preparing for the Research Qualifying Exam. Upon successful completion of the exam, committee members must sign the Research Qualifying Exam Form and submit it to the Graduate Program Administrator.

Preliminary Committee

A student's Research Qualifying Exam is supervised by a preliminary committee consisting of the advisor and at least two additional faculty members.

The preliminary committee is not formally approved by the Office of Graduate Education and does not need to match the student's eventual doctoral committee. However, committee members are typically selected from faculty likely to serve on the doctoral committee.

The committee determines the specific requirements for the exam and evaluates whether the student passes or fails.

Presentation Requirements

The presentation must follow one of the two formats described below:

Option 1: Literature Survey

The student presents a survey of a research area. The presentation should describe the broader field, summarize major contributions and research directions, identify important open problems, and, where appropriate, relate the literature to the student's anticipated research direction.

The presentation should be based on a focused set of papers selected by the student and approved by the committee, typically three to four papers that provide a representative view of the area.

Option 2: Research Paper

The student presents a survey of the research area surrounding a conference or journal paper on which the student is a principal co-author. The paper must be approved by the committee and may be published or under review.

The presentation should place the paper within the broader research landscape by summarizing related work, comparing existing approaches, and explaining the novelty of the student's contribution in the context of prior research.

3.1.4. Doctoral Committee

Each PhD student must form a doctoral committee to oversee the student's dissertation research and administer the Candidacy Exam and Dissertation Defense.

The doctoral committee must consist of at least four members, including the primary advisor and, if applicable, co-advisor(s). Students may elect to include additional committee members.

Committee membership must satisfy the following requirements:

- At least three members must hold tenure or tenure-track appointments in the Department of Computer Science.
- Among these three members, at least one must hold a primary appointment in Computer Science.
- The committee must include one outside member who satisfies the requirements specified in the RPI Catalog.

The doctoral committee must be approved by the Office of Graduate Education before the Candidacy Exam is scheduled. If committee membership changes after the Candidacy Exam, the revised committee must also be approved by the Office of Graduate Education. To nominate and/or change committee membership, students must submit a Nomination of Doctoral Committee Form to the Graduate Program Administrator.

If a committee member retires or otherwise leaves RPI before the student's graduation, that individual may continue to count toward the required Department of Computer Science committee membership, provided that:

- The student graduates within one year of the faculty member's departure; and
- The department head obtains approval from the Office of Graduate Education.

3.1.5. Candidacy Exam

The Candidacy Exam is the third major milestone in the PhD program. It is designed to evaluate the student's dissertation proposal and readiness to complete the proposed research.

To pass the Candidacy Exam, students must:

- Prepare a written dissertation proposal.
- Deliver an oral presentation of the proposal before the doctoral committee.
- Successfully defend the proposal during questioning by the committee.

Full-time students should complete the Candidacy Exam during their third or fourth year and must do so at least one year before the Dissertation Defense. Part-time students may take up to twice as long to satisfy this requirement.

Prior to scheduling the Candidacy Exam, students must have:

- An approved Plan of Study.
- A minimum GPA of 3.0.
- An approved doctoral committee.

Upon successful completion of the exam, students must submit the Record of Candidacy Examination Form to the Graduate Program Administrator.

Written Proposal

The written candidacy proposal should include:

- The proposed research area, including prior work, open challenges, and the expected dissertation contribution.
- Research completed to date.
- Research planned prior to graduation.

There is no required format for the candidacy proposal. However, students are encouraged to follow the formatting guidelines for doctoral dissertations so that material from the proposal may later be incorporated into the final dissertation. Formatting guidance is on the Office of Graduate Education website.

Oral Examination

The Candidacy Exam is administered by the doctoral committee. The student presents the dissertation proposal and is subsequently examined by the committee. Other faculty members may attend the examination but do not participate in the committee's decision.

Students should prepare a presentation of approximately 30–45 minutes, excluding questions. Committee members may ask questions throughout the examination, and the total examination may last up to two hours.

3.1.6. Dissertation and Defense

The Dissertation and Defense is the final milestone in the PhD program. To complete the PhD, students must complete original research, prepare a doctoral dissertation, and successfully defend the dissertation before the doctoral committee.

After passing the Candidacy Exam, students complete the proposed research under the supervision of their advisor and doctoral committee. Students are encouraged to discuss their progress regularly with committee members throughout the dissertation process.

The Dissertation Defense is an oral examination that includes a public presentation of the dissertation research followed by questioning from the doctoral committee. The specific format of the defense is determined by the doctoral committee.

The defense should normally occur at least one year after the Candidacy Exam to allow time for incorporation of feedback received during the candidacy process. Students seeking to defend less than one year after the Candidacy Exam must obtain approval from the Graduate Program Director.

Students are responsible for ensuring that the dissertation complies with Office of Graduate Education requirements. Dissertation formatting guidelines are provided on the Office of Graduate Education website.

Upon successful completion of the defense, students must submit the required dissertation and defense forms to the Graduate Program Administrator.

3.2. Academic Progress

3.2.1. Plan of Study

The Plan of Study is the list of all the courses the student has taken or plans to take to fulfill the requirements for the degree. The Department, the Office of Graduate Education, and the Registrar's Office refer to the Plan of Study when they check whether students are fulfilling the degree requirements. When it comes time for a student to graduate, the Registrar's Office will compare the student’s Plan of Study with their transcript to be certain that the right courses were taken.

Students can fill out the Plan of Study electronically or fill out a printed version. Students will probably have to revise the Plan of Study several times, so it will be easier to make revisions if the form is completed electronically. Always use the most recent Plan of Study. Older/outdated versions will not be accepted.

Students are encouraged to meet with the Computer Science Graduate Program Administrator to prepare a draft of their Plan of Study (POS) before meeting with their advisor.

PhD Plan of Study for students with MS from another school

Students who earned an MS degree from another school should list this degree on their Plan of Study as counting for 24 credits. An additional 48 credits must then be listed to bring the total to 72 for the PhD. The 24 credits for the MS are considered course credits, but they do not count as specific courses. The additional 48 credits must include at least 12 course credits and at least 24 dissertation credits.

Pursuing MS and PhD degrees at RPI

Students pursuing both an MS and a PhD should file separate Plan of Study forms for each degree. In this case, students may list any course credits from an RPI MS on the PhD Plan of Study. MS thesis and project credits do not count as either course or dissertation credits for the PhD. See the Graduate Program Administrator for a detailed process. A new Plan of Study and Change of Status form are required to be approved and on file before proceeding.

3.2.2. Annual Progress Review

At the end of each spring semester, all PhD students must complete a Doctoral Student Yearly Review (DSYR) form documenting progress toward degree requirements and goals for the coming year. The student's advisor reviews the report and evaluates whether progress has been satisfactory. The completed report and updated Plan of Study are then reviewed by the Graduate Program Director.

3.2.3. Milestone Deadlines

Students are expected to make timely progress toward all PhD milestones. Full-time students must satisfy milestone deadlines described in Section 3.1. Part-time students may take up to twice as long to satisfy these requirements.

3.2.4. Extensions

Students who anticipate being unable to meet a milestone deadline must request an extension in advance. Extension requests are reviewed by the Graduate Curriculum Committee.

3.2.5. Unsatisfactory Progress and Dismissal

Students may be subject to dismissal from the PhD program for reasons including:

- Failure to maintain a minimum GPA of 3.0.
- Accumulating two failing course grades.
- Failure to complete program milestones within approved timelines.
- Two consecutive annual reviews indicating unsatisfactory progress.
- Other failures to make satisfactory academic or research progress.

Students entering the PhD program with a prior Master's degree are expected to make progress toward degree milestones at an accelerated pace.

See the RPI catalog for the Graduate Academic Suspension and Dismissal Policy.

4. Colloquium Attendance

Full-time Computer Science graduate students are expected to attend department colloquia throughout their graduate studies. Attendance is formally tracked during the first four semesters of full-time enrollment, during which students must attend at least 50% of the colloquia offered.

Attendance is measured from the student's first semester in the graduate program, even if the student is concurrently enrolled in an undergraduate program. For students in the accelerated BS–PhD program, attendance tracking begins upon completion of 128 undergraduate credits.

Department of Computer Science colloquia count toward this requirement, as well as Computer Science candidacy exams and dissertation defenses. Any other events, such as research group meetings, department seminars, and talks sponsored by other departments, do not count.

Students who fall short of the attendance requirement in a given semester may make up the difference through additional attendance in a subsequent semester. Students with recurring schedule conflicts should contact the Graduate Program Administrator by the end of the semester to discuss possible accommodations.

Part-time students are not required to satisfy the colloquium attendance requirement but are encouraged to attend department colloquia and other research talks whenever possible. Students should consult the Graduate Program Administrator regarding attendance tracking.

Appendix A Course Lists for Core Buckets

Students may petition the GCC to count a special topics course in a Core Bucket with advisor support.

Theory/Algorithms (T):

- CSCI 6220 Randomized Algorithms*
- CSCI 6510 Distributed Systems and Algorithms*
- CSCI 6040 Approximation Algorithms*
- CSCI 6450 Principles of Program Analysis
- CSCI 6100 Machine Learning From Data
- CSCI 6230 Cryptography and Network Security I
- CSCI 6240 Cryptography and Network Security II
- CSCI 6250 Frontiers of Network Science

- CSCI 6560 Computational Geometry
- CSCI 6050 Theory of Computation*
- CSCI 6610 Computing and Quantum Computing
- ECSE 6530 Information Theory and Coding
- MATP 6620 Combinatorial Optimization and Integer Programming

***** course can be used to satisfy Option 3 for Core Foundation: Algorithms.

Special topics courses that have already been approved to satisfy this area are:

- Network Resilience (CSCI)
- Economics and Computation (CSCI)
- Algorithmic Game Theory (CSCI) *
- Quantum Computer Programming (ECSE)
- Network Flows (CSCI)
- Algorithm Design & Modern ML (CSCI) *
- Designing Economic Algorithms (CSCI)
- Distributed Quantum Computing (CSCI)
- Computational Aspects of Online Learning (CSCI)
- Topics in Algorithms (CSCI)

Systems, Software and Security (SSS):

- CSCI 6190 Decentralized AI (AI & Blockchain)
- CSCI 6360 Parallel Computing**
- CSCI 6450 Principles of Program Analysis***
- CSCI 6460 Large-Scale Programming and Testing
- CSCI 6230 Cryptography and Network Security I
- CSCI 6240 Cryptography and Network Security II
- CSCI 6500 Distributed Computing Over the Internet***
- CSCI 6310 Networking in Linux Kernel
- CSCI 6280 Graph Mining
- CSCI 6510 Distributed Systems & Algo
- ECSE 6600 Internet Protocols
- ECSE 6730 Fault-Tolerant Digital Systems

****** course can be used to satisfy Option 3 for Core Foundation: Systems.
*******course can be used to satisfy Option 3 for Core Foundation: Software/Programming Languages.

Special topics courses that have already been approved to satisfy this area are:

- Software Verification (CSCI)
- Security and Privacy of Machine Learning (CSCI)
- Network Security and Defense (CSCI)
- Programming in Haskell (CSCI)
- Quantum Programming and Applications (CSCI)
- Cloud Computing (CSCI)
- Quantum Computer Systems (CSCI)
- Frontiers in Blockchain Research (CSCI)
- Hardware Security (CSCI)
- Quantum Computer Organization (CSCI)

AI/ML/Data (AMD):

- CSCI 6100 Machine Learning From Data
- CSCI 6190 Decentralized AI (AI & Blockchain)
- CSCI 6390 Data Mining
- CSCI 6110 Computational Social Processes
- CSCI 6130 Natural Language Processing
- CSCI 6350 Data Science
- CSCI 6400 Xinformatics
- CSCI 6340 Ontologies
- CSCI 6250 Frontiers of Network Science
- CSCI 6140 Machine Learning and Optimization
- CSCI 6370 Data & Society
- CSCI 6120 Computational Finance
- CSCI 6270 Computational Vision
- CSCI 6160 Reinforcement Learning
- CSCI 6240 Time Series Analysis
- CSCI 6280 Graph Mining
- ECSE 6530 Information Theory and Coding
- ECSE 6610 Pattern Recognition
- ECSE 6850 Intro. To Deep Learning
- ISYE 6100 Time Series Analysis

Special topics courses that have already been approved to satisfy this area are:

- Information Retrieval (COGS)
- Economics and Computation (CSCI)
- Trustworthy Machine Learning (CSCI)

- Machine Learning Bioinformatics (CSCI)
- Safe Autonomy (CSCI)
- Projects in Machine Learning (CSCI)
- Network Resilience (CSCI)
- AI for Conservation (CSCI)
- Machine Learning Seminar (CSCI)
- Deep Learning on Graphs (CSCI)
- Neurosymbolic Knowledge Graphs (CSCI)
- AI for Drug Discovery (CSCI)
- AI for Science (CSCI)
- Large Language Models (CSCI)
- Decision Making under Uncertainty (CSCI)
- Sense Making and Computation (CSCI)
- Deep Learning (CSCI)
- Computational Creativity (ECSE)
- Distributed Optimization and Learning (ECSE)
- Knowledge Graph-Powered Hybrid AI (CSCI)
- Stochastic Optimization and Reinforcement Learning (MATP)
- Frontiers in Blockchain Research (CSCI)
- Computational Aspects of Online Learning (CSCI)
- Intro to Logic-Based AI (CSCI)
- Foundations of LLMs (CSCI)
- Advanced Language Models (CSCI)

Mathematics and Numerical Computing (MNC)

- CSCI 6800 Computational Linear Algebra
- CSCI 6820 Numerical Solution of Ordinary Differential Equations
- CSCI 6840 Numerical Solution of Partial Differential Equations
- CSCI 6860 Finite Element Analysis
- CSCI 6140 Machine Learning and Optimization
- CSCI 6120 Computational Finance
- MANE 6660 Fundamentals of Finite Elements
- MATP 6600 Intro. To Optimization
- MATP 6610 Computational Optimization
- MATP 6620 Combinatorial Optimization and Integer Programming
- MATP 6640 Linear and Conic Optimization
- MATH 6660 Stochastic Methods & Processes

Special topics courses that have already been approved to satisfy this area are:

- Distributed Optimization and Learning (ECSE)

Robotics/Vision/Graphics (RVG)

- CSCI 6490 Robotics II
- CSCI 6270 Computational Vision
- CSCI 6530 Advanced Computer Graphics
- CSCI 6550 Interactive Visualization
- CSCI 6560 Computational Geometry
- ECSE 6470 Robotics I
- ECSE 6650 Computer Vision
- ECSE 6620 Computer Vision for Visual Effects
- ECSE 6630 Digital Image and Video Processing
- ECSE 6640 Digital Picture Processing

Special topics courses that have already been approved to satisfy this area are:

- Safe Autonomy (CSCI)
- AI for Conservation (CSCI)

Appendix B MS Quantum Track Course Lists

Quantum Computing Depth Bucket

- CSCI 6610 - Computing and Quantum Computing
- CSCI 6620 - Quantum Computer Organization

Special topics courses that have already been approved to satisfy this area are:

- CSCI 6963 - Distributed Quantum Computing
- ITWS 6960 - Applied Quantum Computing
- MATH 6890 – Numerical Methods for Quantum Systems
- ECSE 6969 – Quantum Information Processing

Quantum Computing Supplementary Courses

- CSCI 6230 - Cryptography and Network Security
- CSCI 6050 - Theory of Computation
- MATH 4100 - Linear Algebra
- MATH 6660 - Stochastic Processes and Modeling
- MATP 4600 - Probability Theory and Applications
- PHYS 4100 - Introductory Quantum Mechanics

- PHYS 6510 - Quantum Mechanics I